

Spin-Orbit Torque MRAM for ultrafast embedded memories: from fundamentals to large scale technology integration

Kevin Garello
Memory department
imec
Leuven, Belgium
kevin.garello@imec.be

Farrukh Yasin
Memory department
imec
Leuven, Belgium
farrukh.yasin@imec.be

Gouri Sankar Kar
Memory department
imec
Leuven, Belgium
gouri.kar@imec.be

Abstract—Spin-orbit torque (SOT) magnetic random access memory (MRAM) is an emerging non-volatile memory that offers efficient and reliable sub-ns switching. It is considered as a promising candidate to tackle the replacement of SRAM memories in low level caches (L1-3). In this paper, we review the fundamentals of spin-orbit torque and we describe full-scale integration of perpendicular magnetized SOT-MRAM devices on 300mm wafers, using manufacturable methods and CMOS compatible processes. We demonstrate state-of-the-art properties of W-based top-pinned magnetic tunnel junction (MTJ) with perpendicularly magnetic anisotropy (PMA) that possess large endurance ($>10^{11}$), and that can be switched at sub-nanosecond regimes (210ps) with power as low as 130pJ on 60nm devices. We conclude by opening current research focus and recent progress made by our team.

Keywords—MRAM, spin orbit torque, large scale, integration, spin hall effect, MTJ

I. INTRODUCTION

The introduction of non-volatility (NV) at cache level is a major challenge for advance logic nodes as it would lead to a large decrease of the power consumption of microprocessors. Among NV memory technologies, Spin-Transfer-Torque (STT) magnetic random-access memory (MRAM) has gained a lot of attention due to its scalability, low power and relatively low access times, as well as a compatibility with scaled CMOS processes and voltages. In fact, past years have seen major foundries and tool suppliers investing significant R&D resources into embedded STT-MRAM. They recently started prototyping demonstrators, progressively maturing for mass production [1-5]. Standard MRAM devices are magnetic tunnel junctions (MTJ). They consist of two ferromagnetic layers with perpendicular magnetic anisotropy (PMA), separated by a thin oxide tunnel barrier, one of the layers being the storage layer (free layer) and the other used as reference layer (RL) (Fig. 1). Depending on the relative orientation of the magnetization of these two layers (parallel/anti-parallel), the MTJ cell exhibits low/high resistance, measured through the tunnel magneto-resistance effect (TMR), defining the reading state (0/1). The most common writing operation relies on STT, the transfer of spin angular momentum from the reference layer to the free layer. However, STT-MRAM cannot reliably operate at ns and sub-ns scales because of large incubation delays (due to the spin injection symmetry), making it an unsuitable solution to tackle L1-3 SRAM cache replacement and non-volatile logic. In addition, the shared read/write path can impair the read reliability, while the write current can impose a severe stress for the memory cell leading to a possible time dependent degradation of the MTJ. To mitigate these issues, Spin Orbit Torque MRAM (SOT-MRAM) has been proposed [6-8]. SOT-MRAM uses a three terminal MTJ-based concept to isolate the read and the write

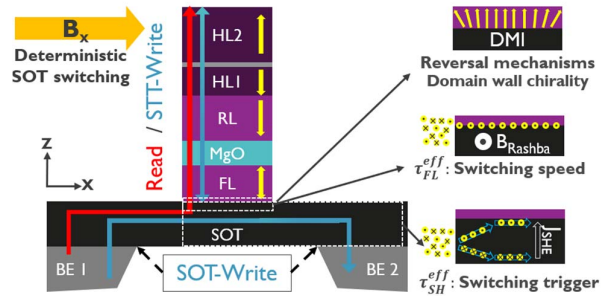


Fig. 1: Schematic of MTJ stack with top-pinned SAF design and following composition: W(SOT)/CoFeB(FL)/MgO/CoFeB(RL)/SAF (HL1-HL2). Compared to STT, SOT writing is decoupled from TMR reading using 3 terminal geometry, which allows larger cell endurance. SOT converts charge into spin current, through Spin Hall and Rashba interaction, while DMI governs reversal mechanisms.

path (Fig. 1), significantly improving device endurance and read stability. Unlike STT, magnetization reversal by spin-orbit torque is performed using in-plane currents. It however requires the assist of an in-plane assist field, applied along the current direction, to ensure deterministic switching [6-8]. Due to the nature of the spin torque injection geometry (perpendicular to easy-axis), incubation time is negligible and allows reliable switching operation at sub-ns timescales [9]. SOTs are characterized by two current induced spin current sources [10, 11] – the physical origin is still debated in literature but can be simplified as follow: a damping-like torque, generally attributed to spin Hall effect (τ_{SH}), and a field-like torque, generally attributed to Rashba interaction (τ_{FL}). Today, understanding is that τ_{SH} triggers switching while τ_{FL} accelerates it. Additionally, a static interaction, named Dzyaloshinskii–Moriya (DMI), has an important role in SOT-induced magnetization switching. Indeed, it cants magnetization at the edges of the sample, which later favors domain wall nucleation and propagation reversal mechanisms [12-14]. SOT-MTJ proof of concept being established in laboratories [15, 16], the next natural step was to prove the feasibility of integration on large scale wafers. This is what we will describe in the following: the first-time integration of SOT-MTJ cells on 300mm wafers using manufacturable processes [17], and associated stack developments. We demonstrate low power sub-ns switching, excellent endurance and absence of electro-migration effect of ultrathin SOT layers. We also present pathways to further reduction in the switching power.

II. SOT-MRAM INTEGRATION

A. SOT and stack optimization

SOT efficiency is described by charge-to-spin conversion coefficients θ_{SH} and θ_{FL} [11]:

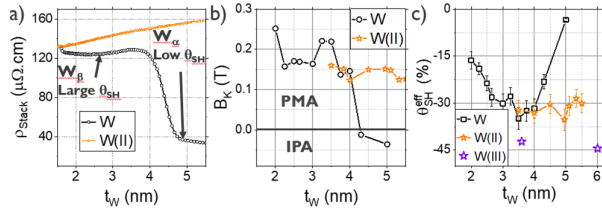


Fig 2: SOT layer optimization as a function of W thickness t_W : a) Stack resistivity, b) Effective anisotropy B_k , c) SOT efficiency. For W (black), we have transition from β to α phase at ~ 4 nm with PMA loss and strong θ_{SH} decrease above 4 nm due to W_α phase, whereas our W(II) process (orange) allows enlarging β -phase window while maintaining large θ_{SH} and PMA.

$$\theta_{SH/FL}^{eff} = \frac{2e}{h} M_s t_{FL} \frac{\tau_{SH/FL}^{eff}}{j_c}, \quad (1)$$

where e , h , M_s , t_{FL} , j_c are respectively the electron charge, the reduced Planck constant, the saturation magnetization of FL, the FL thickness and the charge current density injected into the SOT/FL track. SOTs are usually large in heavy metals, and tungsten (W) in its beta phase has the largest reported efficiency in common metals: $\theta_{SH} \sim -30\%$ [18]. Note that new exotic materials, such as topological insulators (BiSe, BiSe...) [19], are reporting even larger values ($>100\%$), but are to date extremely complex to integrate in fab environment and to combine with pMTJs. As W is commonly used in CMOS technology, it is a natural choice to develop our technology using it. W is however a very specific material that possesses two crystalline phases: an alpha phase W_α that has low resistivity but also low θ_{SH} , and a beta phase W_β with larger resistivities but also very large θ_{SH} . This is shown in Fig. 2.a, where we characterize stack resistivity (ρ_w) as function of W thickness (t_w) for tri-layer stack $W(t_w)/\text{CoFeB}(1\text{nm})/\text{MgO}$, representing the base of a MTJ stack. We identify a phase transition $W_{\alpha-\beta}$ at about 4.2 nm. Additionally, we characterized the t_w dependence of magnetic anisotropy (B_k) and of SOT efficiency using harmonic Hall voltage method [10], this is summarized in Fig. 2 b,c. It is important to note that the oxide process significantly impact all these parameters compared to standard thermal SiO_2 (not shown here). For the simplified stack, we find an optimum of properties at $t_w \sim 4$ nm, with θ_{SH} up to -32% , in line with literature values [18], and $B_k \sim 0.15$ T. Above 4 nm, W turns into alpha phase which dramatically reduces resistivity ($30 \mu\Omega\cdot\text{cm}^2$), θ_{SH} (-2%), but also suppresses the perpendicular anisotropy (PMA), leading to an in-plane magnetized free layer (IPA). This low thickness phase transition $W_{\alpha-\beta}$ is potentially a reliability issue, due to thickness and process variations, SOT efficiency could vary significantly across wafer. We therefore developed a new process for W that allows enlarging β -phase window while maintaining large θ_{SH} and PMA, as shown in Fig. 2 (W(II) process, orange). Today, this process has been even more optimized and allows for $\theta_{SH} > 40\%$. For integration demonstration, we selected $t_w = 3.6$ nm. Subsequently, we developed and stabilized a top-pinned W-MTJ stack SOT/FL/MgO/RL/Ru(0.85)/HL, with optimized SAF (RL/HL) to minimize offset field B_{off} at small device dimensions: SOT= W(3.6), FL= CoFeB(1), RL= CoFeB(1.1)/W(0.3)/Co(0.6), HL= Pt(0.8)/[Co(0.3)Pt(0.8)]₆. After annealing at 300°C, the blanket TMR and resistance area (RA) are measured at 100% and $14 \Omega\cdot\mu\text{m}^2$.

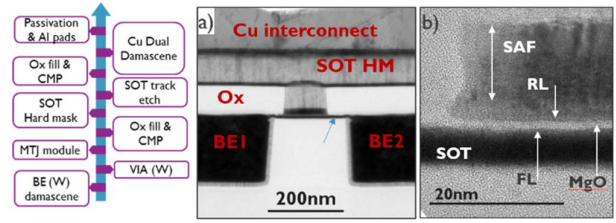


Fig 3: Left chart resumes the main integration modules of SOT-MTJ technology. a) lateral TEM cross section view of a 3-terminal cell showing a MTJ with straight profile sitting on a continuous SOT layer contacting 2 smooth BE, b) High resolution TEM on pillar edge highlighting sidewalls clean of shorts across the MgO barrier and reflecting the excellent stop MTJ etch condition stabilized.

B. 300mm Integration

To integrate SOT-MTJ technology, we designed a SOT-MTJ dedicated mask set, and use imec 300 mm fab MRAM line. The main steps of the integration process are summarized in Fig. 3: bottom electrodes (BE) are fabricated using a smooth tungsten damascene process. Then, the MTJ pillar is defined using 193i nm deep immersion (193i) lithography and etched using ion beam etching at normal and grazing angle. We developed specific stop conditions based on end-point detection to maintain the SOT layer intact while patterning the pillar. As shown on TEM picture Fig. 3.b, these conditions leave the device without sidewall shorts across the MgO barrier. During SOT module, we use a TiN hard mask to define the SOT layer shape with 193i lithography. After oxide opening, the remaining material is ion beam etched. A Cu dual damascene module allows to interconnect top and bottom contacts to metallization pads, completing the electrical connection of a 3T-SOT-MTJ.

III. DEVICE PERFORMANCES

A. SOT-MTJ characteristics

Once integration is completed, we characterize at room temperature the standard MTJ properties as a function of pillar diameter (CD), namely the TMR, the coercivity field B_c and the offset field B_{off} . Additionally, we estimate thermal stability and free layer anisotropy B_k , for P and AP states, using B_c distribution probability method over 350 loops [20]. This is resumed in Fig. 4 for our most up-to-date W-MTJ stack

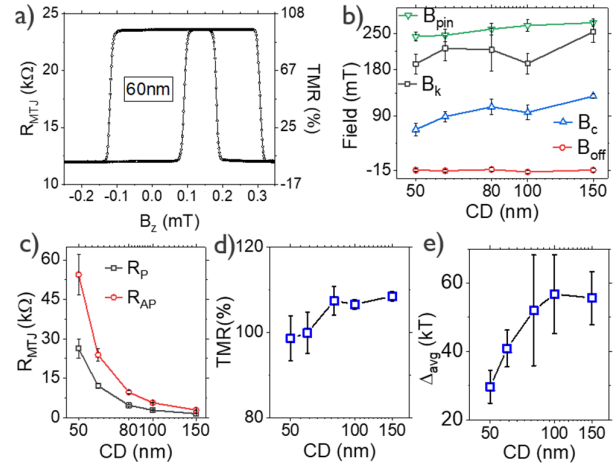


Fig. 4: a) Example of a 60nm R-H loop. MTJ properties vs. size (CD), averaged over 30 devices: b) FL coercivity B_c , anisotropy B_k , offset field B_{off} and RL pinning field B_{pi} , c) Resistances of P and AP states, d) TMR, e) thermal stability.

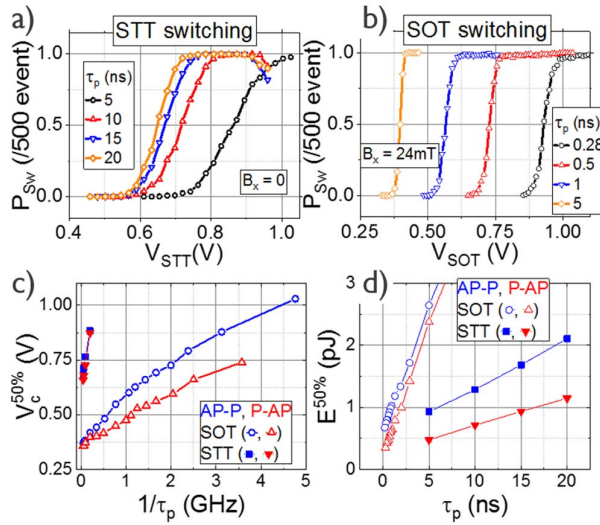


Fig. 5: STT and SOT switching operated separately on *exact same* 62nm first generation device: a,b) examples of switching distribution as a function of pulse voltage $V_{STT,SOT}$ for various pulse length τ_p (P-AP transition), c) $V_c^{50\%}$ vs. inverse of pulse length showing switching up to 5 GHz. Remark that the offset field generates an imbalance of $V_{c,SOT}^{50\%}$ up to 15% between P-AP and AP-P transitions, d) Energy vs. time showing ultrafast and low power SOT-device performances.

version. As an example, we measure for a 60 nm device: a TMR of 100 %, $B_k \sim 190$ mT, $B_c \sim 90$ mT, $B_{off} \sim 15$ mT, a pinning field $B_{pin} \sim 250$ mT and $\Delta \sim 50$. It shows that our W_β -MTJ has already very good properties, similar to state-of-art top-pinned stacks. Using on-wafer 4 probe test structures, we quantify post process SOT resistivity, and find $\rho_{SOT} \sim 160 \mu\Omega \cdot \text{cm}$ ($\rho_{Stack} \sim 140 \mu\Omega \cdot \text{cm}$), which confirms the excellent stop etch conditions stabilized for the MTJ pillar patterning.

B. Electrical performances

In this section, we discuss initial results obtained on devices of eCD = 62 nm, sitting on a 210 nm wide SOT track ($R_{SOT} = 300 \Omega$) [14]. Using the *exactly same device*, we can evaluate STT and SOT switching distribution probability P_{sw} for 500 events average as a function of voltage pulse length and amplitude (Fig. 5a, b). The critical switching voltage reported is evaluated at 50% probability. While STT case is measured at 0 external field, SOT switching is characterized for an in-plane field $B_x = 23$ mT. In STT case, τ_p is limited to 5ns to avoid breakdown. $V_{c,STT}^{50\%}$ is ~ 1 V to switch both P and AP states, corresponding to an energy of ~ 470 fJ at 5ns.

In contrast, SOT switching is not limited by pulse duration, and similar to best results shown in literature [14, 16], we report reliable switching down to 210 ps, with $V_{c,SOT}^{50\%} \sim 1$ V, and 0.7 V at 500 ps. Although the voltage is reasonably low, the current density injected in the SOT track is extremely large ($\sim 300 \text{ MA} \cdot \text{cm}^{-2}$ at 280ps). However, SOT switching is energy wise very favorable with a value as low as 350 fJ at 280 ps. This is due to the current injection geometry and to the low resistance of SOT track. Further reduction in energy at sub-ns can be obtained by current density reduction via design optimization, stack engineering and mostly using more efficient SOT materials.

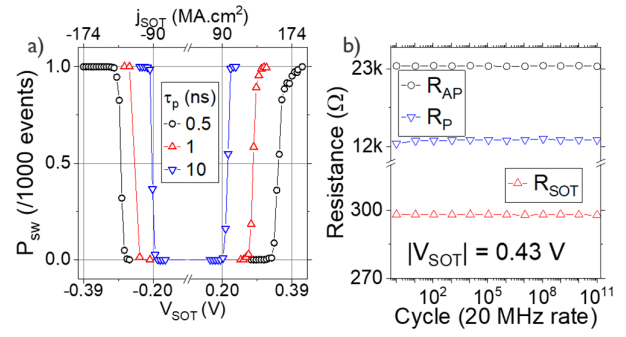


Fig. 6: New device generation. a) Switching probability measurement at 0.5, 1 and 10 ns showing significant improvement of voltage and current density switching, b) Endurance measured for 10^{11} cycles at a repetition rate of 20 MHz with following conditions: $\tau_p = 1$ ns and $|V_p| = 0.43$ V. No SOT/TMR resistance degradation is observed despite large stress applied on the cell.

C. Toward cell improvement

Since our initial results [17], stack has been improved, layout has been re-designed to incorporate BE to SOT vias and dense dummification structures with single pillar connection ($260 \times 540 \text{ nm}^2$ pitch, $110 \text{ nm} \times 110 \text{ nm}$ gap at design). This allow us evaluating impact of etch on future arrays. The results are extremely positive, and we have a large device yield all across the 300mm wafers. All these progress and optimizations conducted to an improvement of device performances, such as switching current density and writing energy reduction. As illustration, we show in Fig. 6a a switching probability distribution for a 60 nm device. At 1 (0.5) ns pulse length, we record an average (P-AP and AP-P) current density of 112 (135) $\text{MA} \cdot \text{cm}^{-2}$, corresponding to an energy of 205 (130) fJ. We also report that despite the large stress applied to the SOT track, our devices are very robust. This is highlighted by a test endurance measurement of 10^{11} events using 1 ns / $|130| \text{ MA} \cdot \text{cm}^{-2}$ pulses with a repetition rate of 20 MHz (Fig. 6.b) - neither SOT nor TMR degradation is observed and switching operation voltage are weakly affected.

IV. CONCLUSION

Across this review paper, we demonstrated how we fabricated state-of-the-art SOT-MRAM devices using CMOS-compatible large-scale integration processes. We discussed the stack optimization and the main integration steps. We also showed the possibility to improve the technology by SOT layer and MTJ stack engineering, as well as layout optimization. The sub-ns switching speed and the low power device performances show that SOT-MRAM has the capacity to tackle SRAM cache replacement, but improvements are still required. This will be the focus of our future research: reducing the current density and obtaining manufacturable field-free switching solutions via new SOT material development and device design.

REFERENCES

- [1] Y. K. Lee et. al, "Embedded STT-MRAM in 28-nm FDSOI Logic Process for Industrial MCU/IoT Application", IEEE Symposium on VLSI Technology, pp.181-182 (2018)
- [2] Y. C. Shih et. al, "A 1-Mb 28-nm 1T1MTJ STT-MRAM With Single-Cap Offset-Cancelled Sense Amplifier and In Situ Self-Write-Termination", IEEE Journal of Solid-State Circuits **54**, pp.231-239 (2019)

- [3] K. Lee et. al, "22-nm FD-SOI Embedded MRAM with Full Solder Reflow Compatibility and Enhanced Magnetic Immunity", Symposium on VLSI Technology, pp.183-184 (2018)
- [4] L. Xue et. al, "Process Optimization of Perpendicular Magnetic Tunnel Junction Arrays for Last-Level Cache beyond 7 nm Node" IEEE Symposium on VLSI Technology, pp.117-118 (2018)
- [5] O. Golonzka et. al, "MRAM as Embedded Non-Volatile Memory Solution for 22FFL FinFET Technology", IEEE IEDM, p. 18-1 (2018)
- [6] I. M. Miron et. al, "Perpendicular switching of a single ferromagnetic layer induced by in-plane current injection", *Nature* **476**, 189–193 (2011)
- [7] L. Liu, C-F Pai, Y. Li, H. W. Tseng, D. C. Ralph and R. A. Buhrman, "Spin-Torque Switching with the Giant Spin Hall Effect of Tantalum", *Science* **336**, 555–558 (2012)
- [8] G. Prenat et. al, "Ultra-Fast and High-Reliability SOT-MRAM: From Cache Replacement to Normally-Off Computing", *IEEE Trans. on Multi-Scale Computing Systems*, **2**, 49-60 (2016)
- [9] K. Garello et. al, "Ultrafast magnetization switching by spin-orbit torques", *Appl. Phys. Lett.* **105** (21), 212402
- [10] K. Garello et. al, "Symmetry and magnitude of spin-orbit torques in ferromagnetic heterostructures", *Nature nanotechnology* **8**, 587 (2013)
- [11] A. Manchon, I. M. Miron, T. Jungwirth, J. Sinova, J. Zelezny, A. Thiaville, K. Garello and P. Gambardella, "Current-induced spin-orbit torques in ferromagnetic and antiferromagnetic systems", arXiv:1801.09636, in press
- [12] N. Mikuszeit, O. Boulle, I. M. Miron, K. Garello, P. Gambardella, G. Gaudin, L. D. Buda-Prejbeanu, "Spin-orbit torque driven chiral magnetization reversal in ultrathin nanostructures", *PRB* **92**, 144424 (2015)
- [13] E. Martinez, Luis Torres, N. Perez, M. A. Hernandez, V. Raposo and S. Moretti, "Universal chiral-triggered magnetization switching in confined nanodots", *Scientific Reports* **5**, 10156 (2015)
- [14] M. Baumgartner et. al, "Spatially and time-resolved magnetization dynamics driven by spin-orbit torques", *Nat. Nanotech.* **12**, 980–986 (2017)
- [15] M. Cubucku et. al, "Spin-orbit torque magnetization switching of a three-terminal perpendicular magnetic tunnel junction", *Appl. Phys. Lett.* **104**, 042406 (2014)
- [16] M. Cubucku et. al, "Ultra-fast perpendicular spin-orbit torque MRAM", *IEEE Transactions on Magnetics* **54**, 1-4 (2018)
- [17] K. Garello et. al, "SOT-MRAM 300mm integration for low power and ultrafast embedded memories", 2018 IEEE Symposium on VLSI Circuits, 81-82 (2018)
- [18] C-F Pai, L. Liu, Y. Li, H.W. Tseng, D.C. Ralph, R.A. Buhrman, "Spin transfer torque devices utilizing the giant spin Hall effect of tungsten", *App. Phys. Lett.*, **101**, 122404 (2012)
- [19] C-F Pai, "Switching by topological insulators", *Nature Materials* **17**, p. 755–757 (2018)
- [20] L. Thomas et. al, "Perpendicular spin transfer torque magnetic random access memories with high spin torque efficiency and thermal stability for embedded applications", *Journal of Appl. Phys.* **115**, 172615 (2014)